

# Generation Interconnection System Impact Study Report

## For Queue Project Q17-0841

Hermiston 100 kV

July 25, 2019

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0841. The Interconnection Customer proposes a 14.4 MW generating facility interconnecting at the Hermiston 100 kV substation in the Illinois area. The request was received on August 03, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 263 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$653,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                             | Contingency                        | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------------------------------|------------------------------------|-------------|--------------|----------|
| KENDALLVILLE – FARMERSVILLE 100kV                              | JONES CHAPEL – DEKALB 500kV        | 141.4       | 252          | 5.3      |
| HILLCREST – SEGS 5 100kV                                       | RICHLAND – DEKALB 500kV            | 126.4       | 354          | 7.2      |
| 43RD ST TERMINAL – JOLIET 29 230kV                             | PUEBLO XTRA – BOCA RATON 500kV     | 125.2       | 816          | 5.4      |
| CUBA CITY – EGE 100kV                                          | JONES CHAPEL – DEKALB 500kV Line   | 123.0       | 251          | 5.3      |
| LANDSMAN CREEK – 24TH & GARDENDALE 230kV                       | RICHLAND – DEKALB 500kV Line       | 122.8       | 589          | 9.1      |
| SIDNEY A MURRAY JR HYDRELECTRIC – 24TH & GARDENDALE 230kV Line | RICHLAND – DEKALB 500kV line       | 121.7       | 615          | 9.1      |
| MINNEHAHA – NAPOLEON PEAKING STATION 230kV L                   | TROY ENERGY LLC – RICHLAND 500kV   | 118.0       | 977          | 13.0     |
| BLUEBONNET – RUSSELL FORK 230kV                                | JONES CHAPEL – DEKALB 500kV        | 115.4       | 1,597        | 32.0     |
| CONCORD – UNIVERSITY 230kV Line 1                              | CONCORD – UNIVERSITY 230kV         | 114.4       | 767          | 12.1     |
| EGE – GOLDEN GATE 100kV                                        | DEKALB – JONES CHAPEL 500kV        | 111.7       | 265          | 5.3      |
| STATION S VAN NUYS – GOULD 230kV                               | TROY ENERGY LLC – RICHLAND 500kV L | 111.5       | 632          | 6.6      |
| CONCORD – UNIVERSITY 230kV line                                | UNIVERSITY – CONCORD 230kV line    | 110.6       | 793          | 12.1     |
| ADIRONDACK 500 kV/230 kV Autotransformer                       | E METHUEN 500/230 kV Auto          | 109.5       | 1,026        | 5.1      |
| ST. LAWRENCE – JAMES DE YOUNG 230kV                            | LONETREE – E METHUEN 500kV         | 109.1       | 1,026        | 6.9      |

| Monitored Facility                     | Contingency                                              | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------|----------------------------------------------------------|-------------|--------------|----------|
| JONES CHAPEL – DEKALB 500kV            | RUSSELL FORK – ROBISON PARK 230kV Line                   | 108.5       | 3,605        | 66.7     |
| DEKALB 230/1 kV Auto                   | RICHLAND – DEKALB 500kV L                                | 107.9       | 2,093        | 39.0     |
| DEKALB 500-1 kV Transformer            | DEKALB – RICHLAND 500kV                                  | 107.9       | 2,093        | 39.0     |
| MADELINE – EAST METHUEN 230kV          | EAST METHUEN – GRAY'S FLAT 230kV                         | 107.9       | 1,074        | 9.5      |
| EASTOWN RD – SWAN VALLEY 230kV         | SWAN VALLEY – TREGO 500kV                                | 107.1       | 819          | 6.3      |
| TWIN COUNTY RANCH – RICHLAND 230kV     | RICHLAND – TROY ENERGY LLC 500kV                         | 106.2       | 1,190        | 19.4     |
| TREGO 500/1kV Transformer              | TREGO – TROY ENERGY LLC 500kV                            | 105.8       | 769          | 12.9     |
| SABANERA SUBSTATION – LONETREE 500kV L | MONTPELIER ELECTRIC GENERATING STA – MUSKOGEE 500kV Line | 105.6       | 2,866        | 10.8     |
| TWIN COUNTY RANCH – SWAN VLY 230kV     | TROY ENERGY LLC – RICHLAND 500kV                         | 105.3       | 946          | 17.3     |
| WINHROP – HEBO 230kV line              | BARRETT – HEBO 230kV                                     | 105.0       | 629          | 5.5      |
| MUSKOGEE – DARLINGTON ROAD 100kV Line  | DEKALB – RICHLAND 500kV Line                             | 104.9       | 427          | 6.7      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                                       | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| KENDALLVILLE – FARMERSVILLE 100kV                              | Rebuild line with higher-capacity conductor | \$5,222,000   | 0% (\$6,000)          |
| SEGS 5 – HILLCREST 100kV                                       | Rebuild line with higher-capacity conductor | \$4,273,000   | 0% (\$5,000)          |
| JOLIET 29 – 4RD ST TERMINAL 230kV                              | Rebuild line with higher-capacity conductor | \$4,011,000   | 0% (\$3,000)          |
| CUBA CITY – EGE 100kV                                          | Rebuild line with higher-capacity conductor | \$6,170,000   | 0% (\$7,000)          |
| 24TH & GARDENDALE; – LANDSMAN CRK 230kV Line                   | Rebuild line with higher-capacity conductor | \$45,205,000  | 0% (\$52,000)         |
| SIDNEY A MURRAY JR HYDRELECTRIC – 24TH & GARDENDALE 230kV Line | Rebuild line with higher-capacity conductor | \$17,255,000  | 0% (\$20,000)         |
| NAPOLEON PEAKING STA – MINNEHAHA 230kV                         | Rebuild line with higher-capacity conductor | \$29,774,000  | 0% (\$34,000)         |
| RUSSELL FORK – BLUEBONNET 230kV                                | Rebuild line with higher-capacity conductor | \$82,880,000  | 0% (\$93,000)         |
| CONCORD – UNIVERSITY 230kV Line 1                              | Rebuild line with higher-capacity conductor | \$41,009,000  | 0% (\$44,000)         |

| Facility                                 | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------|---------------------------------------------|---------------|-----------------------|
| EGE – GOLDEN GATE 100kV                  | Rebuild line with higher-capacity conductor | \$4,063,000   | 0% (\$5,000)          |
| STATION S VAN NUYS – GOULD 230kV         | Rebuild line with higher-capacity conductor | \$12,962,000  | 0% (\$15,000)         |
| CONCORD – UNIVERSITY 230kV line          | Rebuild line with higher-capacity conductor | \$60,225,000  | 0% (\$65,000)         |
| ADIRONDACK 500 kV/230 kV Autotransformer | Replace transformer with higher-rated unit  | \$25,888,000  | 0% (\$29,000)         |
| ST. LAWRENCE – JAMES DE YOUNG 230kV      | Reconductor line                            | \$23,058,000  | 0% (\$26,000)         |
| JONES CHAPEL – DEKALB 500kV              | Reconductor line                            | \$49,966,000  | 0% (\$58,000)         |
| DEKALB 230/1 kV Auto                     | Replace transformer with higher-rated unit  | \$6,580,000   | 0% (\$8,000)          |
| DEKALB 500-1 kV Transformer              | Replace transformer with higher-rated unit  | \$26,536,000  | 0% (\$32,000)         |
| E METHUEN – MADELINE 230kV               | Reconductor line                            | \$10,295,000  | 0% (\$12,000)         |
| EASTOWN ROAD – SWA VALLEY 230kV          | Reconductor line                            | \$3,146,000   | 0% (\$4,000)          |
| TWIN COUNTY RANCH – RICHLAND 230kV       | Reconductor line                            | \$11,877,000  | 0% (\$14,000)         |
| TREGO 500/1kV Transformer                | Replace transformer with higher-rated unit  | \$23,362,000  | 0% (\$27,000)         |
| LONETREE – SABANERA 500kV Line           | Reconductor line                            | \$64,187,000  | 0% (\$72,000)         |
| SWAN VLY – TWIN COUNTY RANCH 230kV line  | Reconductor line                            | \$10,286,000  | 0% (\$12,000)         |
| WINHROP – HEBO 230kV line                | Reconductor line                            | \$3,577,000   | 0% (\$4,000)          |
| MUSKOGEE – DARLINGTON ROAD 100kV Line    | Reconductor line                            | \$4,982,000   | 0% (\$6,000)          |
|                                          | Total allocated to project                  |               | \$653,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.